3 ASSIGNMENTS

Take-Home Review

01 **Campaign Response Analysis**
PA + Life cross-sell, dashboard

02 **Agent Contract Schema**
Monthly KPI validation, salary ⇄ commission

03 **InsureX Sales Agent**
Agentic RAG + lead collection

Data analysis · Dashboard



PA and Life Are Not the Same Customer

PA sells on how often the account is used, not wealth. Life sells on income, savings and tenure with the bank.

1.23%
2,581 of 210,458

0.76%
PA

0.46%
Life

EXHIBIT 2 — **The Apr–Sep rise is targeting, not season**
Pool was 85% active vs 29% the rest of the year · χ^2 p<0.001

EXHIBIT 3 — **Lower Mass leads PA; Upper Mass leads Life**
Two products, two target lists

EXHIBIT 4 — **Men accept PA at nearly 2× women**
0.98% vs 0.60%; Life flat. Not an age/income proxy · p<0.001

dsc_test_case.csv, cleaned · 210,458 of 215,993 rows

Data analysis · Dashboard

Who Buys What, and What Targeting Works

Insurance Campaign Response

210,458 customers · PA and Life insurance cross-sell campaign

OVERVIEW | DEEP DIVE

EXHIBIT 5

PA and Life Are Opposite Customer Profiles



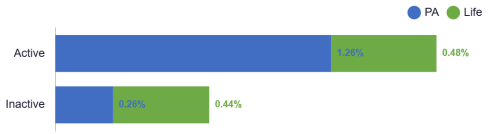
AGE (YRS)	INCOME (K THB)
34y PA 32y Life	11,294 PA 15,140 Life
Nearly identical — not a differentiator here	*Life is 34% higher than PA*

SAVINGS (×100 THB)	TENURE (MO)
896 PA 1,437 Life	39mo PA 54mo Life
Life is 60% higher than PA	*Life is 38% higher than PA*

Median values; Income in thousands and Savings in hundreds of THB so all four axes land on one shared scale. **Life skews toward higher income, more savings and longer tenure; PA toward newer accounts**, with almost no age gap — not the same "buyer," so a campaign needs two target lists, not one.

EXHIBIT 6

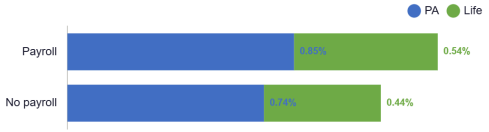
Active Accounts Convert 2.5× Higher — but Only on PA



Active = any inflow/outflow in 30 days (50% of the base). **Active accepts at 1.75% vs. 0.71% inactive** — almost all PA (1.26% vs. 0.26%, ~5×); Life barely moves. Active alone holds 83% of all PA converters.

EXHIBIT 7

Payroll Customers Convert Higher on Both Products



Payroll = salary paid into this bank (21% of the base). **Accepts at 1.40% vs. 1.18%** — moves both PA and Life together (p = 0.017, 0.006), unlike activity above — a secondary signal, not a primary one.

01 PA → active accounts first
Keeps 83% of PA buyers from half the list

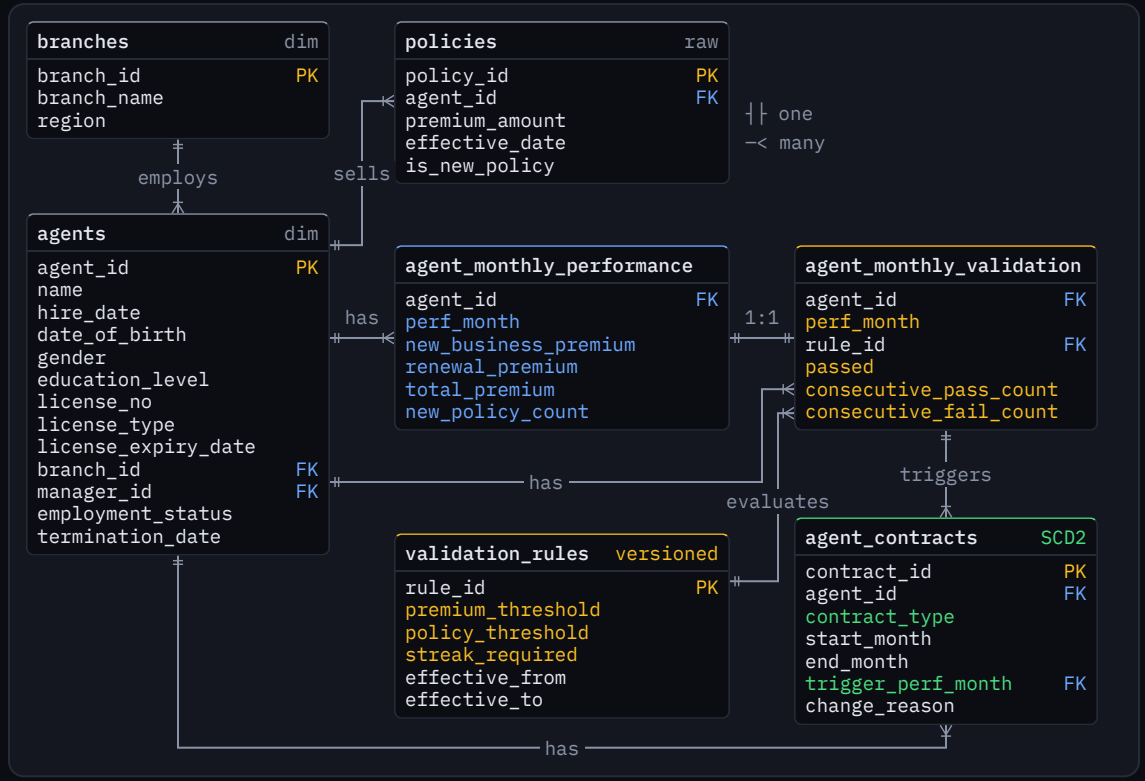
02 Separate PA and Life lists
Different income, savings, tenure profiles

03 Test gender for PA
Men 0.98% vs women 0.60%

Schema design

InsureX Agent Contract Schema

Monthly KPI validation that automatically switches an agent between a salary-based and commission-based contract.



VALIDATION RULE · MONTHLY

`total_premium > 15,000 AND new_policy_count > 5 → PASS`

3 fails in a row → commission-based

3 passes in a row → salary-based

WHY THIS ISN'T THE MINIMUM

Versioned rules

A KPI change is a new row in `validation_rules`, not a migration.

Append-only audit trail

Every contract change points at the exact validation record that triggered it.

New vs. renewal split

Separates the mandatory "new client" gate from proof the book is sticking.

Predictive risk signal

Flags a fail streak before the month closes, not after.

Grounded in research

Checked against Thai licensing, persistency, and commission-chargeback practice, not guessed.

03

InsureX Sales Agent

A LangGraph agent behind a FastAPI REST API that answers insurance product questions via agentic RAG and collects sales leads mid-conversation.

LangGraph | FastAPI · SSE | ChromaDB | SQLite | gpt-4o-mini

narusrn/insurex-sales-agent

The Problem

Answer a customer's product questions accurately, without guessing when the knowledge base doesn't have the answer. When they signal buying interest, collect their contact details.

01 ANSWER
Agentic RAG over 5 Thai insurance product PDFs

02 COLLECT
Name, occupation, income, phone, from the conversation

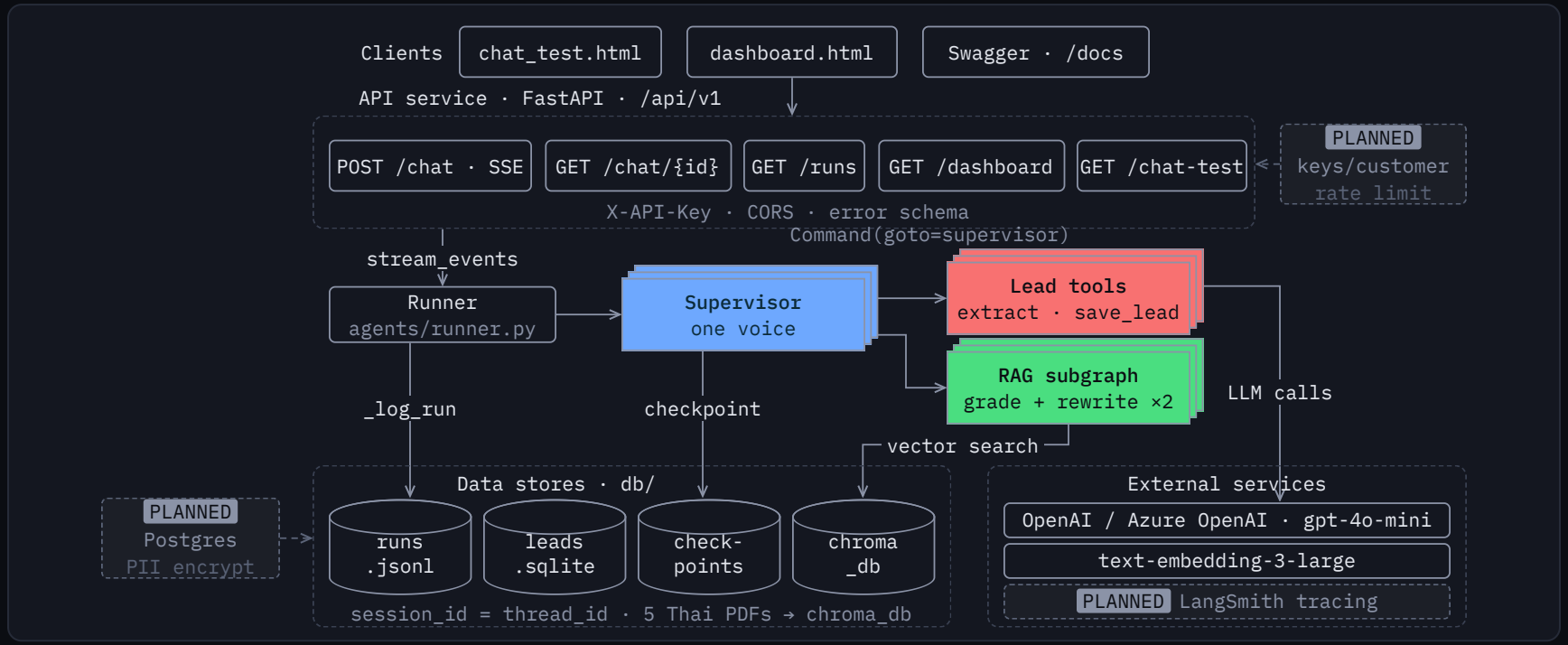
03 GATE
Never save incomplete or unconfirmed personal data

04 ISOLATE
One customer's conversation never leaks into another's

Graded on RAG precision, workflow logic, data integrity, and code quality.

Code quality · Workflow logic

System Architecture Overview



Workflow logic · Code quality

Agentic AI Design Patterns

HAPPY PATH

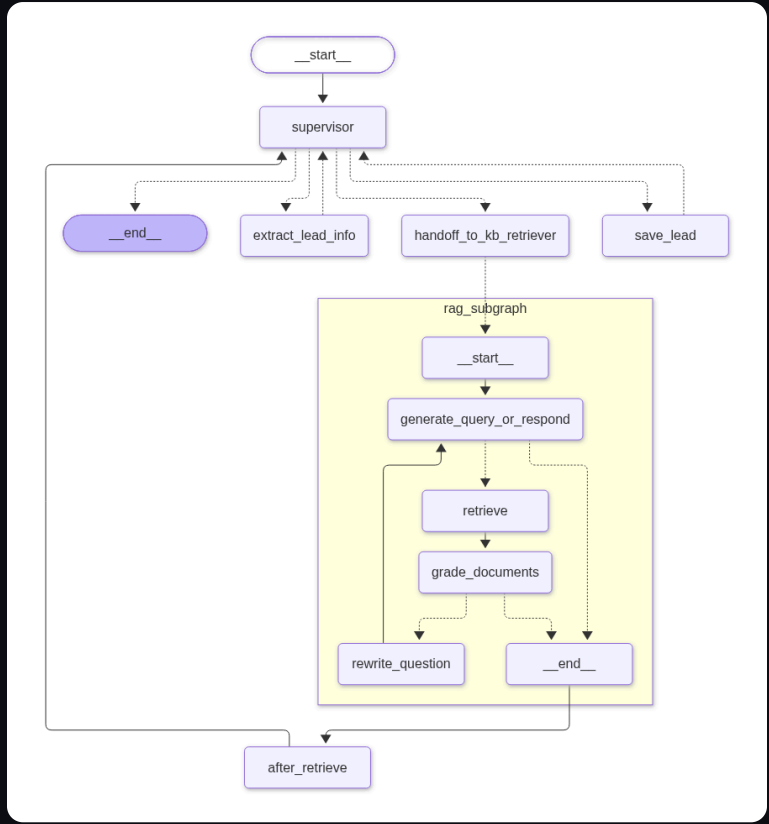
supervisor → rag_subgraph → extract_lead_info → save_lead, each returning to supervisor

ROUTING

Every tool sets `Command(goto=…)`. No static edge leaves a tool node: one was found firing in addition to the Command's own goto.

ONE VOICE

supervisor is the only node that composes a user-facing reply, on every path.



Rendered from the compiled graph · python -m agents.visualize

RAG precision

No Hallucinated Certainty

The grader's verdict, `last_retrieval_relevant`, is shared with the parent graph so supervisor can tell "here's what I found" from "retries ran out."

grade_documents
yes / no against the original question

→

rewrite × 2
RETRIEVER_MAX_REWRITES

→

after_retrieve
still irrelevant → note prefixed ⚠️

→

supervisor
"ไม่พบข้อมูลที่ชัดเจน"

THE BUG, FOUND DURING EVAL

supervisor stated a specific, confident, wrong number pulled from snippets that had just been graded irrelevant.

THE RULE IN _supervisor.md

A ⚠️ note may never be used to state a specific number or condition. Re-verified on all 21 cases, zero regressions.

Data integrity

Won't Save Data It Hasn't Confirmed

01 Completeness

All 4 required fields present and non-blank, or it returns `"Missing: X"`

02 Confirmation

The first complete call never persists. Only `confirmed=True` on a later call, after an explicit "yes".

WHY Every field is Optional

A required argument pressures the model to fabricate a value. The real gate lives in the tool body.

lead-06 · correction, not confirmation · PASS

turn 1	ผมชื่อสมชาย ทำงานช่างไฟฟ้า เงินเดือน 25,000 สนใจ Health Plus ครับ เบอร์ 089-111-2222
Lead Saver	Data complete but not yet confirmed (..., phone=089-111-2222). Restate...
Supervisor	ข้อมูลที่บันทึกไว้มีดังนี้: ... เบอร์โทร: 089-111-2222 ... กรุณายืนยัน...
turn 2	เบอร์ผิดครับ ที่จริงคือ 089-333-4444
Lead Extractor	`{..."phone":"089-333-4444",...}`
Supervisor	ข้อมูลที่บันทึกไว้มีการแก้ไขเบอร์โทรเป็นดังนี้: ... เบอร์โทร: 089-333-4444 ... กรุณายืนยัน...
turn 3	ใช่ครับ ถูกต้องแล้ว
Lead Saver	`Lead saved (id=3)`
DB row	`('สมชาย', 'ช่างไฟฟ้า', '25,000', '089-333-4444', 'Health Plus')`

All four criteria

Eval Harness: Three Separate Layers

21 golden cases · python -m eval.run_full_eval · each layer answers a different question, not one blended score.

01 Pass / fail

Did each case behave as specified?

16/21

eval/demo_log.txt

lead_cases		6/6
session_cases		2/2
qa (strict)		8/13

4 of 5 qa failures (qa-06 · 09 · 11 · 12) are the honest fallback firing correctly

02 Retrieval accuracy

Did it find the right document?

0.91

Hit Rate@k · eval/metrics.py

0.86 MRR

11 qa_cases with a known source doc. Isolates retrieval from generation.

03 LLM-as-judge

Is the reply faithful to what was actually retrieved that run?

11/12

groundedness · eval/judge.py · new

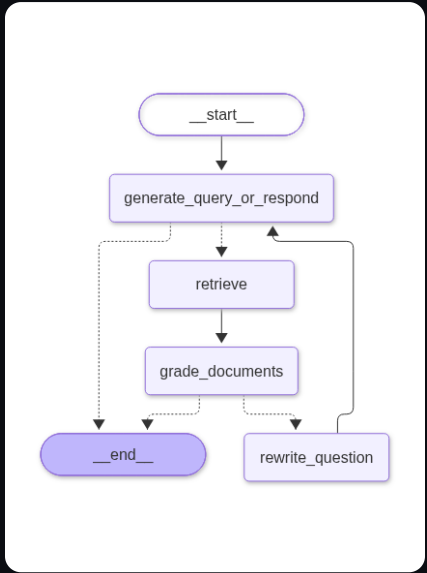
THE ONE MISS · qa-02

Read the premium for age 35 off the "31–40" bracket row. Correct, but not explicitly stated for 35.

Kept as-is, not loosened. Unrelated to its strict-keypoint miss in layer 01.

RAG precision · Workflow logic

The Agentic RAG Retriever



retrieve · grade · rewrite

ChromaDB, text-embedding-3-large, top-k = 2, chunk 500 / overlap 50, max 2 rewrites

qa-06 · eval/demo_log.txt · verbatim

Query	ลูกค้าท้องแล้วเคลมค่าคลอดได้ไหม
Supervisor	ต้องการข้อมูลเกี่ยวกับการเคลมค่าคลอดในประกัน → ค้นข้อมูลในฐานความรู้
RAG Agent	ค้นข้อมูลด้วยคำค้น "เคลมค่าคลอด" · ค้นพบ 2 ข้อความจากฐานความรู้
ตรวจสอบ	ข้อมูลที่ได้ไม่เกี่ยวข้องเพียงพอ ❌ — ลองใหม่ (1/2)
RAG Agent	เขียนคำถามใหม่ · ค้นข้อมูลด้วยคำค้น "ประกันสุขภาพค่าคลอด"
ตรวจสอบ	ข้อมูลที่ได้ไม่เกี่ยวข้องเพียงพอ ❌ — ลองใหม่ (2/2)
RAG Agent	ค้นข้อมูลด้วยคำค้น "ประกันสุขภาพค่าคลอดสำหรับผู้ตั้งครรภ์"
ตรวจสอบ	ข้อมูลที่ได้ไม่เกี่ยวข้องเพียงพอ ❌ — ลองใหม่ (2/2)
Supervisor	ไม่พบข้อมูลที่ชัดเจนเกี่ยวกับการเคลมค่าคลอดในประกัน หากต้องการข้อมูลที่แน่นอน แนะนำให้ติดต่อเจ้าหน้าที่ของ InsureX...

Data integrity · Workflow logic

A Real Conversation, Start to Finish

fda66787-98d8-… · live capture, db/runs.jsonl

Supervisor	สวัสดีครับ! มีอะไรให้ช่วยเหลือเกี่ยวกับประกันหรือผลิตภัณฑ์ของ InsureX ไหมครับ?
RAG Agent	ค้นข้อมูลด้วยคำค้น "แผนประกัน" -- พบ 2 ข้อความ, เกี่ยวข้อง ✅
Supervisor	เรามีแผนประกันหลักๆ ดังนี้ครับ: 1. InsureX Life Secure... 2. InsureX Health Plus...
Extractor #1	{"name":null,"occupation":null,"income":null,"phone":null, "interested_product":"InsureX Health Plus", ...}
Supervisor	ขอข้อมูลเพิ่มเติมหน่อยนะครับ: 1. ชื่อ 2. อาชีพ 3. รายได้ 4. เบอร์โทร
Extractor #2	{"name":"สมชาย ใจดี","occupation":"วิศวกรซอฟต์แวร์", "income":"45,000 บาทต่อเดือน","phone":null, ...}
Supervisor	... -- ยังขาดเบอร์โทรศัพท์ครับ
Lead Saver	Data complete but not yet confirmed (... phone=0812345678). Restate and ask to confirm.
Supervisor	ข้อมูลที่เรามีตอนนี้คือ: ... กรุณายืนยันข้อมูลเหล่านี้ว่าถูกต้องหรือไม่ครับ?
Lead Saver	Lead saved (id=5)

session-01 / 02 · both PASS

ISOLATION

[sess-A] หนูชื่อสมหญิง สนใจ Health Plus ค่ะ

[sess-B] ชื่อฉันคืออะไร

→ ขอโทษครับ ฉันไม่สามารถบอกชื่อของคุณได้ในตอนนี้...

RECALL, SAME SESSION

[sess-C] Health Plus มีแผนอะไรบ้าง → แผนที่ถูกที่สุดคืออะไร

→ แผนที่ถูกที่สุดของประกันสุขภาพ InsureX Health Plus คือ แผน Bronze

session_id = thread_id, AsyncSqliteSaver. Verified against checkpoints.sqlite, not only the reply text.

Code quality

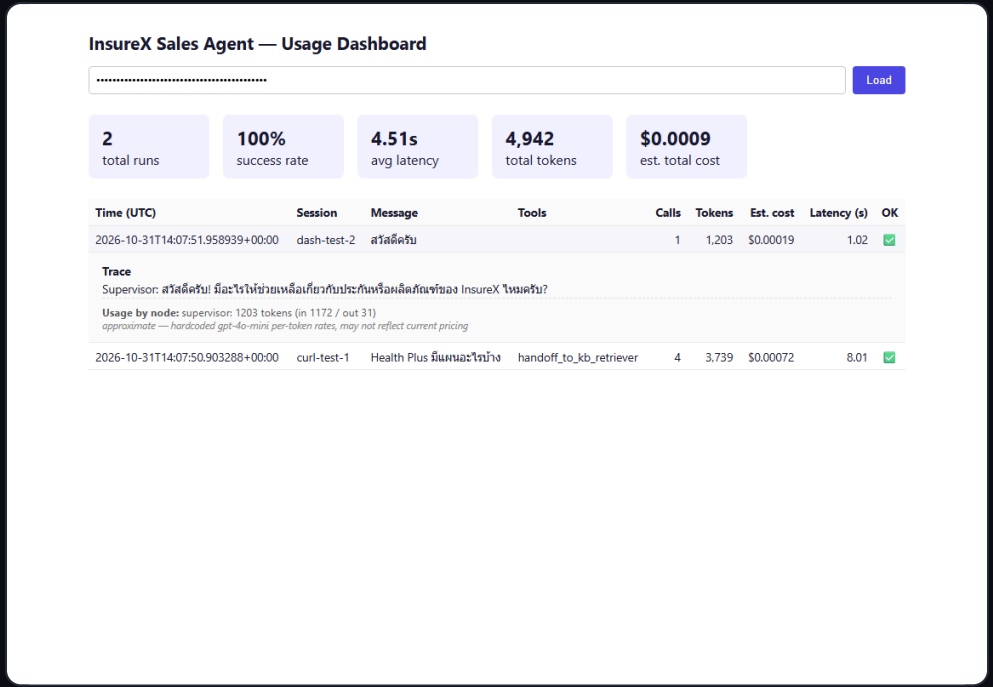
It Runs, With Real Usage Data

db/runs.jsonl · final turn of slide 9's session


```
{
  "message": "ยืนยันครับ ถูกต้อง",
  "tools_used": ["save_lead"],
  "latency": 2.75,
  "llm_calls": 2,
  "usage": {
    "input_tokens": 5123,
    "output_tokens": 129,
    "total_tokens": 5252
  },
  "estimated_cost_usd": 0.000846,
  "cost_note": "approximate -- hardcoded…"
}
```


Summed from usage_metadata on every LLM call in the turn, including those inside rag_subgraph and extract_lead_info.

GET /api/v1/dashboard · expanded row



Code quality

Two Ways to Drive It

`GET /api/v1/chat-test`

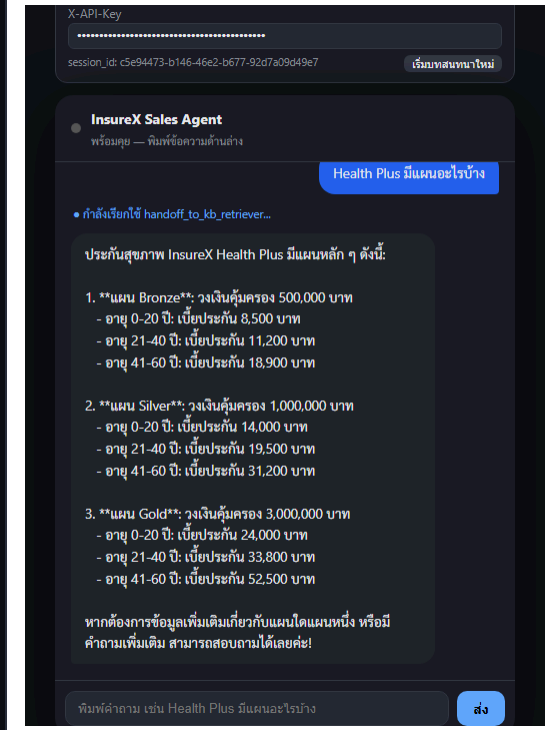
Self-contained HTML page, real token-by-token SSE via fetch + ReadableStream

`GET /api/v1/docs`

Swagger, standardized against narusrn/support-ticket-triage-service

`/api/v1` Prefix on every real route and the docs	`lifespan` One shared checkpointer for the app's lifetime
`{"error":{code,message}}` Same shape on every 4xx/5xx	`GET /health` Outside the version prefix

An earlier Streamlit app was built, then removed in favor of this static page.



Judgment

What I'd Change for Production

01 **Real per-token pricing**
An API lookup or a maintained table with a freshness check, instead of the static _PRICING_PER_1M_TOKENS

02 **Protect lead PII**
Name, phone, income sit in db/leads.sqlite as plaintext, behind one shared X-API-Key

03 **SQLite to Postgres**
Once concurrent write load justifies it; nothing in the schema assumes SQLite

04 **Per-customer keys and rate limits**
Instead of one shared demo key

05 **A real tracing backend**
Point db/runs.jsonl at e.g. LangSmith once volume justifies it

Scope Delivered

WHAT THE BRIEF ASKED FOR

- Agentic RAG over insurance product docs
- Lead extraction and persistence
- Per-session memory and isolation
- Streaming API, demo / test cases

BUILT BEYOND IT, ON REQUEST

- Usage dashboard
- Live chat test page, replacing Streamlit
- Per-turn cost and trace logging
- API layer standardized to a reference project

github.com/narusrn/insurex-sales-agent

Details in README.md